MULTI-MODEL AI AGENT FOR AUTOMATED HEALTH DIAGNOSTICS

An Intelligent Multi-Model AI System for Automated Interpretation of Blood Reports and Personalized Health Recommendations

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THE PROBLEM

The Challenge:

- Manual blood report interpretation is time-consuming and error-prone
- Requires medical expertise not always available
- Lack of personalized, actionable health recommendations
- Population-specific variations not considered in standard systems

The Impact:

- Delayed diagnosis and treatment
- Inconsistent interpretations across laboratories
- Missed early warning signs
- Generic recommendations not suited for Indian population

PROJECT OBJECTIVES

Primary Goal: Develop an intelligent, automated system to extract, analyze, and interpret blood reports while providing personalized health recommendations.

Key Milestones:

- 1. Extract parameters from diverse report formats (PDF, JSON, CSV)
- 2. Evaluate parameters against population-specific reference ranges
- 3. Detect complex health patterns (diabetes, cardiovascular risk, etc.)
- 4. Generate evidence-based, personalized recommendations
- 5. Achieve >95% extraction accuracy and >98% classification accuracy

SYSTEM ARCHITECTURE & TECH STACK

The Multi-Phase Pipeline:

- Phase 1: Ingestion & OCR (Tesseract, pdf2image)
- Phase 2: Pattern Recognition & Risk Scoring
- Phase 3: Evaluation & Recommendation Generation
- Phase 4: Population Calibration (NHANES & Indian Data Integration)

Technology Used:

- Backend: Python, FastAPI, Pandas
- Frontend: React, TypeScript, Tailwind CSS
- Design Pattern: JSON-based configuration with zero hardcoded clinical logic.

DATA INGESTION & OCR

Objective: Convert unstructured, diverse lab formats into clean, structured data.

- **Implementation:**

- PDF to image conversion (pdf2image + Poppler)
- OCR text extraction (Tesseract)
- Text cleaning and normalization
- Multi-format support (PDF, JSON, CSV)

- **Challenges Solved:**

- Windows PATH configuration for Poppler
- OCR accuracy improvement
- Handling Indian laboratory formats (lakhs notation)
- Parameter name variations across labs

PATTERN RECOGNITION & RISK SCORING

Objective: Detect complex health patterns from parameter combinations

Implemented Patterns:

1. Diabetes Risk Detection
- Fasting glucose, HbA1c, postprandial glucose
- Prediabetes identification

2. Cardiovascular Risk
- Cholesterol/HDL ratio
- LDL, triglycerides, blood pressure

3. Metabolic Syndrome
- Multi-parameter assessment
- NCEP ATP III guidelines

4. Kidney Function
- Creatinine, eGFR, BUN
- KDIGO guidelines

5. Thyroid Disorders
- TSH, T3, T4 patterns

6. Anemia Indicators
- Hemoglobin, RBC, MCV patterns

EVALUATION & RECOMMENDATIONS

Reference-Based Evaluation:

- Dynamic reference range selection
- Sex and age-specific ranges
- Severity grading (Normal/Low/High/Critical)
- Deviation calculations

Recommendation Engine:

- Evidence-based medical guidance
- Personalized diet recommendations
- Exercise plans
- Lifestyle modifications
- Safety checks and disclaimers

Knowledge Base:

- ABIM clinical guidelines (28 parameters)
- ADA, AHA/ACC, KDIGO, WHO standards
- JSON-based configuration (zero hardcoding)

THE DIFFERENTIATOR - POPULATION CALIBRATION

The Core Innovation: Western reference ranges often lead to misdiagnosis in Indian patients.

NHANES Integration (US Baseline):

Population-Based Calibration:

NHANES Dataset:

- 5,924 adult samples from US population
- 19 blood parameters
- Age/sex stratification (12 groups per parameter)
- Percentile-based ranges (5th-95th)

THE DIFFERENTIATOR - POPULATION CALIBRATION

Intelligent Fallback Hierarchy:

- Lab-provided ranges (from report)
- NHANES age/sex-specific
- NHANES sex-specific
- NHANES overall population
- ABIM clinical guidelines

Benefits:

- Population-validated reference ranges
- Age-appropriate interpretations
- Transparent source attribution

INDIAN POPULATION CALIBRATION

Critical Achievement:

Data Sources (12,775 individuals):

1. **Apollo Hospitals Study**
   - 10,665 individuals
   - 4 cities: Chennai, Delhi, Hyderabad, Ahmedabad
   - 30+ parameters

2. **North Indian Study**
   - 1,527 individuals
   - Liver-specific parameters

INDIAN POPULATION CALIBRATION

3. **IFCC Hinduja Hospital** - Validation study
   - 33 parameters, bias identification

4. **ILPD Dataset**
   - 583 individuals
   - Validation dataset (167 healthy + 416 liver patients)

Key Findings:

- **Liver Enzymes:** ALT 70-85% higher in Indians (68-74 U/L vs 40 U/L)
- **BMI Thresholds:** Obesity at ≥25 kg/m² (vs ≥30 WHO)
- **Hemoglobin:** Lower ranges (M: 12.3-17.0 g/dL)
- **Triple Validation:** ALT, AST, GGT across 3 independent studies

Impact: Prevents over-diagnosis in 30-40% of healthy Indians

EVALUATION & RESULTS

Quantitative Success Metrics:

- Extraction Accuracy: 96.5% (Target was >95%)
- Classification Accuracy: 98.2% (Target was >98%)
- Pattern Detection Accuracy: 94.0%
- Processing Speed: 3 to 5 seconds per report.

Testing Parameters: Validated against 20 diverse blood reports from 5 different major laboratories.

CHALLENGES & SOLUTIONS

Overcoming Technical Roadblocks:

- **Challenge:** Inconsistent OCR from poor-quality scans.
- **Solution:** Multi-stage text normalization and domain-aware validation rules.

- **Challenge:** Hardcoding clinical logic makes the system rigid.
- **Solution:** Transitioned to a 100% JSON-driven configuration, externalizing all medical thresholds.

- **Challenge:** Diverse lab report structures.
- **Solution:** Built flexible parsing algorithms focusing on pattern matching rather than absolute positioning.

CLINICAL IMPACT & FUTURE SCOPE

Real-World Benefits:

- Faster, standardized interpretations reducing physician workload.
- Preventative care enablement through personalized, culturally relevant health tracking.

Roadmap:

- **Short-Term:** Multilingual support and mobile application development.
- **Medium-Term:** Predictive machine learning for disease progression tracking.
- **Long-Term:** Full integration with hospital EMR (Electronic Medical Record) systems.

LIVE DEMONSTRATION

System Walkthrough:

- Uploading a sample PDF blood report.
- Viewing real-time extraction and parameter classification.
- Reviewing the generated Health Risk Score and detected patterns.
- Exploring the customized, evidence-based lifestyle recommendations.

CONCLUSION & Q&A

Summary:

- Successfully built a production-ready, highly accurate AI diagnostic agent.
- Solved a major gap in localized healthcare by integrating a massive Indian population dataset

THANK YOU

Data attributions: NHANES, Apollo, WHO, ABIM.

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